Fractions and decimals — Rounding · Multiplying and dividing · Recurring decimals · Mixed numbers

No calculator. Give fractions in their simplest form. ★ = challenge question.

A · DECIMALS

1 Round to 2 decimal places:

a 47.826 **b** 3.1049 **c** $0.096\,51$

2 Round to 3 decimal places:

a 8.12345 **b** $0.045\,678$ **c** 219.9996

3 Use a written method to calculate **a** 3.4×2.6 **b** 14.5×12.3

4 Calculate **a** $0.5208 \div 0.004$ **b** $71.4 \div 0.03$

5 Write each recurring decimal using dot notation:

a $0.5555\ldots$ **b** $0.454545\ldots$ **c** $0.217217\ldots$

6 Write each fraction as a decimal (use dot notation if it recurs):

a $\frac{4}{9}$ **b** $\frac{7}{12}$ **c** $\frac{3}{8}$ **d** $\frac{5}{11}$

B · FRACTIONS

7 Calculate

a $\frac{2}{5} \times \frac{10}{11}$ **b** $-\frac{3}{8} \times \frac{4}{9}$ **c** $\frac{7}{12} \times \left(-\frac{3}{14}\right)$ **d** $-\frac{5}{6} \times \left(-\frac{9}{10}\right)$

8 Evaluate

a $\frac{1}{4} \div \frac{1}{3}$ **b** $\frac{3}{5} \div \frac{1}{10}$ **c** $-\frac{2}{3} \div \frac{5}{9}$ **d** $-\frac{7}{8} \div \left(-\frac{7}{12}\right)$

9 Alex ran $1\frac{3}{4}$ miles, swam $\frac{2}{3}$ of a mile and cycled $4\frac{5}{6}$ miles. How far did he travel in total?

10 Calculate

a $2\frac{3}{5} + 3\frac{1}{4}$ **b** $8\frac{2}{3} + 1\frac{5}{6}$ **c** $5\frac{1}{2} - 2\frac{4}{5}$ **d** $4\frac{1}{8} - 1\frac{5}{6}$

11 Evaluate, giving your answers in their simplest form: **a** $1\frac{1}{2} \times 2\frac{2}{5}$ **b** $2\frac{2}{3} \times 1\frac{1}{8}$

12 Write each recurring decimal as a fraction: **a** $0.\dot{6}$ **b** $0.\dot{3}\dot{6}$ **c** $0.5\dot{2}$

13 Find the side length of a cube with volume **a** $\frac{8}{125}$ m³ **b** $\frac{1}{64}$ cm³

Check: count decimal places when multiplying · make the divisor whole when dividing · mixed numbers → improper fractions before × or ÷ · simplest form at the end.